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M150 PS 1

1 a)

continuous exponential growth model $\Rightarrow N(t) = N(0)e^{rt}$

$r = ?$

given $\rightarrow$ $N(0) = 1$

$N(5) = N(0)e^{r \cdot 5} = e \cdot N(0)$ "increasing every 5 days by a factor of e"

$\rightarrow e^{5r} = e^1$

$\ln(\cancel{e}^{5r}) = \cancel{\ln}(\cancel{e}^1)$

$5r = 1$

$r = \frac{1}{5}$

$N(t) = 1 \cdot e^{\frac{1}{5}t}$

1 b)

of new cases appearing on day $t = \frac{dN}{dt}$ (instantaneous rate of change)

$$\frac{dN}{dt} = \frac{d}{dt}\left(e^{t/5}\right) = \frac{1}{5}e^{t/5}$$

$$\frac{dN/dt}{N(t)} = \frac{\frac{1}{5}e^{t/5}}{e^{t/5}} = \frac{\cancel{e^{t/5}}}{5} \times \frac{1}{\cancel{e^{t/5}}} = \frac{1}{5} \Rightarrow 0.2 \text{ per day}$$

1 c)

$$R_I(t) = \left.\frac{dN}{dt}\right|_{t-3} = \frac{1}{5} e^{\frac{t-3}{5}} = \frac{1}{5}\left(e^{\frac{t}{5}} \cdot e^{\frac{-3}{5}}\right)$$

$$* \quad \frac{1}{5} e^{t/5} = \frac{dN}{dt}$$

$$e^{\frac{-3}{5}} \cdot \frac{dN}{dt}$$

$$0.549 \cdot \frac{dN}{dt}$$

reported # of positive tests on day t is about 54.9% of the actual new cases on day t b/c of the 3 day lag

2a)

$$\frac{dN(t)}{dt} = \frac{rN(t)}{1-e^{-N(t)/N(T)}}$$

when $N(t) << N(T)$

$x = N(t)/N(T) << 1$

$\hookrightarrow$ Taylor expand e^{-x} for small x

$$e^{-x} = 1 - x + \frac{x^2}{2} - \frac{x^3}{6} - \dots$$

$$\hookrightarrow 1 - e^{-x} = 1 - \left(1 - x + \frac{x^2}{2} - \frac{x^3}{6} - \dots\right)$$

$$= x - \frac{x^2}{2} + \frac{x^3}{6} + \dots \approx x$$

small $x \Rightarrow 1 - e^{-x} \approx x = N(t)/N(T)$

so: $$\frac{dN}{dt} = \frac{rN(t)}{N(t)/N(T)} = \frac{r\cancel{N(t)}}{1} \cdot \frac{N(T)}{\cancel{N(t)}} = r \cdot N(T)$$

at early times where $N(t) << N(T)$

$$\frac{dN}{dt} = r \cdot N(T) \Rightarrow \text{constant}$$

$$\int_{N(0)}^{N(t)} dN' = \int_0^t r \cdot N(T)\, dt'$$

$$\boxed{\text{at early times, population growth is linear}}$$

$$N(t) - N(0) = r \cdot N(T) \cdot t$$

$$N(t) = N(0) + r \cdot N(T) \cdot t$$

2b)

$$\frac{dN(t)}{dt} = \frac{rN(t)}{1 - e^{-N(t)/N(T)}}$$

when $N(t) \gg N(T)$

$x = N(t)/N(T) \gg 1$ $\rightarrow$ $\frac{1}{e^{\text{big\#}}} \Rightarrow \frac{1}{e^{\infty}} \Rightarrow 0$

so:

$$e^{-N(t)/N(T)} \Rightarrow 0$$

so:

$$1 - e^{-N(t)/N(T)} = 1 - 0 = 1$$

so ODE:

$$\frac{dN}{dt} = \frac{rN(t)}{1}$$

$$\int \frac{dN}{N(t)} = \int r\,dt$$

$$e^{\ln(N(t))} = e^{(rt + c)}$$

$$N(t) = e^{rt} \cdot e^{c}$$

$$e^{c} = N_0^*$$

$\rightarrow$ checking if $N(0) = N_0^*$

$\rightarrow N(0) = N_0^* \cdot e^{r \cdot 0}$

$= N_0^* \cdot 1$

$= N_0^*$ ✓

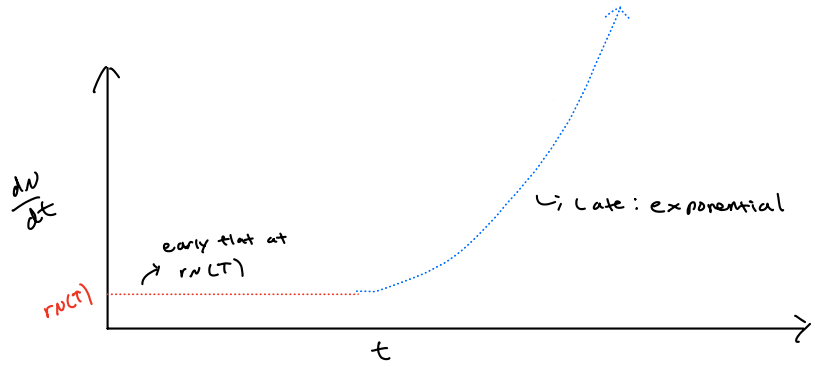
$$N(t) = N_0^* \cdot e^{rt}$$

at long times, population grows exponentially w/ rate r

2c)

Early: $(N(t) << N(T)) \simeq$ constant

Late: $(N(t) >> N(T)) \simeq$ exponential



2d)

At early times, $N(t) = N(0) + rN(T) \cdot t$ is already a linear function. That being said, the original linear space of N vs. t is already the best space possible. No transformation is needed.

3a)

exponential growth

$\hookrightarrow \Delta N(t) = ae^{bt}$ $\quad$ a = initial value, b = growth rate

$\frac{\Delta N(t+\Delta t)}{\Delta N(t)}$ = constant

$\hookrightarrow \frac{ae^{b(t+\Delta t)}}{ae^{bt}} = e^{b\Delta t}$

$\frac{\Delta N(10)}{\Delta N(5)} = \frac{2800}{7} \cong 7$ $\quad$ $\frac{\Delta N(15)}{\Delta N(10)} \cong \frac{8800}{2800} \cong 3.1$

$\frac{\Delta N(20)}{\Delta N(15)} \cong \frac{20,000}{8800} \cong 2.3$ $\quad$ $\frac{\Delta N(25)}{\Delta N(20)} \cong \frac{27,000}{20,000} \cong 1.35$

$\frac{\Delta N(30)}{\Delta N(25)} \cong \frac{43000}{27000} \cong 1.6$ $\quad$ $\frac{\Delta N(35)}{\Delta N(30)} \cong \frac{33000}{43000} \cong 0.77$

Ratio is not constant, decreases over time dropping below 1. New cases per day is not exponential.

Early on, data does look exponential, but after day 25 curve bends over. Growth rate peaks around day 29 and then declines. Cumulative total flattens, suggesting not exponential growth and curve S-shaped w/ some K carrying capacity.

Approx. Diff Eq. $\Rightarrow$ logistic equation

solving diff. eq. that would better describe the data

$$\frac{dN}{dt} = rN\left(1-\frac{N}{K}\right) \Rightarrow \frac{dN}{N(1-\frac{N}{K})} = r\,dt$$

$$\Rightarrow \frac{1}{N(1-\frac{N}{K})} = \frac{A}{N} + \frac{B}{1-\frac{N}{K}} \Rightarrow 1 = A\left(1-\frac{N}{K}\right) + BN$$

$$\Rightarrow N=0: \quad A=1$$

$$N=K: \quad BK = 1 \Rightarrow B = \frac{1}{K}$$

$$\Rightarrow \frac{1}{N(1-\frac{N}{K})} = \frac{1}{N} + \frac{1/K}{1-N/K}$$

$$\int\left(\frac{1}{N} + \frac{1/K}{1-N/K}\right)dN = \int r\,dt$$

$$u = 1-\frac{N}{K}, \quad du = -\frac{1}{K}\,dN$$

$$\int \frac{1/u}{u}(-u\,du) = -\int\frac{du}{u} = -\ln|u|$$

$$\ln|N| - \ln\left|1-\frac{N}{K}\right| = rt + C_1$$

$$\ln\left(\frac{N}{1-\frac{N}{K}}\right) = \ln\left(\frac{NK}{K-N}\right) = \ln(K) + \ln\left(\frac{N}{K-N}\right)$$

constant $\rightarrow C$

$$\ln\left(\frac{N}{K-N}\right) = rt + C$$

$$e^{\ln\left(\frac{N}{N-K}\right)} = e^{rt + \ln\left(\frac{N_0}{K-N_0}\right)}$$

$$\frac{N}{N-K} = e^{rt}\cdot e^{\ln\left(\frac{N_0}{K-N_0}\right)}$$

$$\frac{N}{N-K} = e^{rt}\cdot\left(\frac{N_0}{K-N_0}\right)$$

$* \; N(0) = N_0, \; t=0$

$$\ln\left(\frac{N_0}{K-N_0}\right) = C$$

letting $A = \frac{N_0}{K - N_0}$

$$\frac{N}{N-K} = e^{rt} \cdot A$$

$$\frac{1}{A} = \frac{K - N_0}{N_0}$$

$$N = Ae^{rt} \cdot (N - K)$$

$$N = NAe^{rt} - KAe^{rt}$$

$$N - NAe^{rt} = -KAe^{rt}$$

$$N(1 - Ae^{rt}) = -KAe^{rt}$$

$$N = \frac{-KAe^{rt}}{1 - Ae^{rt}} = \frac{KAe^{rt}}{1 + Ae^{rt}}$$

dividing numerator and denominator by Ae^{rt}

$$N = \frac{K}{\frac{1}{A}e^{-rt} + 1}$$

$$N(t) = \frac{K}{1 + \left(\frac{K - N_0}{N_0}\right)e^{-rt}}$$

Testing Capacity:

checking for exponential growth:

$$\frac{T(14)}{T(7)} = \frac{33000}{6000} = 5.5 \qquad \frac{T(21)}{T(14)} = \frac{99000}{33000} = 3$$

$$\frac{T(28)}{T(21)} = \frac{120000}{99000} = 1.21 \qquad \frac{T(35)}{T(28)} = \frac{138000}{120000} = 1.15$$

- not exponential since ratio not constant
- testing capacity is also growing about linear later

$$\frac{dT}{dt} = c_T \Rightarrow T(t) = T(0) + c_T \cdot t$$

Comparing growth rates
(testing capacity vs. newly confirmed cases):

Day	New cases/day	tests/day	Ratio
10	2747	12,000	0.23
15	8821	33,000	0.27
20	19,965	87,000	0.23
25	27,043	96,000	0.28
30	43,438	111,000	0.39
35	33,288	138,000	0.24
40	29,164	147,000	0.20

- ratio of confirmed cases vs. tests processed stays around 0.20 ~ 0.30
- since the confirmed cases are a near constant fraction of tests performed, the # of cases confirmed is mainly driven by testing capacity

3b)

Reported = New infections × P(symptoms) × P(tested | symptoms)

= 2,000,000 × 0.20 × 0.50

= 200,000

200,000 new cases per day

3c)

$$\text{Positives} = \text{Tests per day} \times P(\text{Covid} \mid \text{resp. symp})$$

$$= 150{,}000 \times 0.30$$

$$= 45{,}000 \text{ new cases reported per day}$$

NO, with only 150,000 tests / day the reported positives are capped at 45,000 / day regardless of whether its growing exponentially or is flat (linear).

That being said, the 150,000 / day makes the the two scenarios indistinguishable.